

# 1. Sensor

## 1. 공정 split 평가 분석 - 설비 센서 파라미터

| lot_id | wafer_id | Avg(precursor... | Avg(chamber... | Avg(rf_power)... | Max(reflect_p... |
|--------|----------|------------------|----------------|------------------|------------------|
| AAJ.1  | 13       | 29.16            | 490.83         | 1124.39          | 4.37             |
| AAJ.1  | 14       | 29.17            | 491.54         | 1125.24          | 5.35             |
| AAJ.1  | 15       | 29.17            | 491.16         | 1125.06          | 5.10             |
| AAJ.1  | 16       | 29.17            | 491.79         | 1125.04          | 4.38             |
| AAJ.1  | 17       | 29.17            | 491.22         | 1124.90          | 4.89             |
| AAJ.1  | 18       | 29.17            | 491.78         | 1125.36          | 5.35             |
| AAJ.1  | 19       | 29.17            | 491.44         | 1124.55          | 5.83             |
| AAJ.1  | 20       | 29.16            | 491.09         | 1124.93          | 322.50           |
| AAJ.1  | 21       | 29.17            | 490.91         | 1124.95          | 4.83             |
| AAJ.1  | 22       | 29.17            | 491.33         | 1124.71          | 5.43             |
| AAJ.1  | 23       | 29.17            | 491.43         | 1124.46          | 5.65             |
| AAJ.1  | 24       | 29.17            | 492.02         | 1125.23          | 5.71             |
| AAJ.1  | 25       | 29.17            | 491.92         | 1124.98          | 5.09             |
| AAK.1  | 1        | 29.17            | 492.15         | 1124.72          | 5.34             |
| AAK.1  | 2        | 29.17            | 492.86         | 1125.47          | 295.42           |
| AAK.1  | 3        | 29.17            | 491.68         | 1124.67          | 5.89             |
| AAK.1  | 4        | 29.17            | 492.13         | 1124.78          | 5.26             |
| AAK.1  | 5        | 29.16            | 491.63         | 1125.35          | 4.93             |
| AAK.1  | 6        | 29.17            | 491.83         | 1125.59          | 5.13             |
| AAK.1  | 7        | 29.17            | 492.02         | 1125.02          | 5.44             |
| AAK.1  | 8        | 29.17            | 490.39         | 1124.39          | 4.93             |
| AAK.1  | 9        | 29.16            | 491.47         | 1125.13          | 4.86             |
| AAK.1  | 10       | 29.17            | 492.03         | 1125.07          | 6.08             |
| AAK.1  | 11       | 29.16            | 491.57         | 1125.38          | 5.55             |
| AAK.1  | 12       | 29.16            | 491.31         | 1124.84          | 284.24           |
| AAK.1  | 13       | 29.17            | 492.10         | 1124.80          | 325.05           |
| AAK.1  | 14       | 29.16            | 491.29         | 1125.21          | 5.03             |
| AAK.1  | 15       | 29.17            | 492.73         | 1125.08          | 5.19             |
| AAK.1  | 16       | 29.17            | 491.65         | 1125.25          | 4.93             |
| AAK.1  | 17       | 29.17            | 491.77         | 1124.89          | 233.39           |
| AAK.1  | 18       | 29.16            | 491.60         | 1124.97          | 5.22             |
| AAK.1  | 19       | 29.17            | 491.74         | 1124.84          | 4.72             |
| AAK.1  | 20       | 29.16            | 491.41         | 1124.93          | 223.35           |
| AAK.1  | 21       | 29.16            | 491.77         | 1125.39          | 5.15             |

## 2. Defect

### 1. 공정 split 평가 분석 - defect

| lot_id | wafer_id | Sum(value) for |
|--------|----------|----------------|
| AAA.1  | 1        | 610.00         |
| AAA.1  | 2        | 709.00         |
| AAA.1  | 3        | 717.00         |
| AAA.1  | 4        | 9051.00        |
| AAA.1  | 5        | 714.00         |
| AAA.1  | 6        | 6106.00        |
| AAA.1  | 7        | 710.00         |
| AAA.1  | 8        | 719.00         |
| AAA.1  | 9        | 8268.00        |
| AAA.1  | 10       | 713.00         |
| AAA.1  | 11       | 725.00         |
| AAA.1  | 12       | 706.00         |
| AAA.1  | 13       | 704.00         |
| AAA.1  | 14       | 719.00         |
| AAA.1  | 15       | 711.00         |
| AAA.1  | 16       | 711.00         |
| AAA.1  | 17       | 722.00         |
| AAA.1  | 18       | 704.00         |
| AAA.1  | 19       | 722.00         |
| AAA.1  | 20       | 8849.00        |
| AAA.1  | 21       | 6502.00        |
| AAA.1  | 22       | 715.00         |
| AAA.1  | 23       | 721.00         |
| AAA.1  | 24       | 727.00         |
| AAA.1  | 25       | 717.00         |
| AAB.1  | 1        | 717.00         |
| AAB.1  | 2        | 711.00         |
| AAB.1  | 3        | 726.00         |
| AAB.1  | 4        | 721.00         |
| AAB.1  | 5        | 718.00         |
| AAB.1  | 6        | 722.00         |
| AAB.1  | 7        | 710.00         |
| AAB.1  | 8        | 715.00         |
| AAB.1  | 9        | 715.00         |

Data table:

1. 공정 split 평가 ...

3. Total Data

1. 공정 split 평가 분석 - 설비 센서 파라미터 (2)

| lot_id | wafer_id | Avg(precursor... | Avg(chamber... | Avg(rf_power)... | Max(reflect_p... | Sum(value) for |
|--------|----------|------------------|----------------|------------------|------------------|----------------|
| AAA.1  | 1        | 29.17            | 491.59         | 1125.19          | 5.20             | 610.00         |
| AAA.1  | 2        | 29.17            | 492.31         | 1125.02          | 4.87             | 709.00         |
| AAA.1  | 3        | 29.17            | 491.78         | 1125.14          | 5.59             | 717.00         |
| AAA.1  | 4        | 29.16            | 490.91         | 1125.32          | 341.92           | 9051.00        |
| AAA.1  | 5        | 29.17            | 491.70         | 1124.23          | 5.39             | 714.00         |
| AAA.1  | 6        | 29.17            | 491.52         | 1123.97          | 259.20           | 6106.00        |
| AAA.1  | 7        | 29.17            | 492.37         | 1125.66          | 4.63             | 710.00         |
| AAA.1  | 8        | 29.17            | 490.81         | 1125.12          | 5.20             | 719.00         |
| AAA.1  | 9        | 29.16            | 490.29         | 1125.02          | 353.93           | 8268.00        |
| AAA.1  | 10       | 29.17            | 492.09         | 1124.60          | 4.63             | 713.00         |
| AAA.1  | 11       | 29.17            | 491.52         | 1125.00          | 5.21             | 725.00         |
| AAA.1  | 12       | 29.16            | 491.65         | 1125.11          | 5.25             | 706.00         |
| AAA.1  | 13       | 29.17            | 491.41         | 1125.00          | 5.12             | 704.00         |
| AAA.1  | 14       | 29.16            | 492.02         | 1125.07          | 5.05             | 719.00         |
| AAA.1  | 15       | 29.17            | 492.13         | 1125.09          | 5.69             | 711.00         |
| AAA.1  | 16       | 29.17            | 490.89         | 1124.88          | 5.66             | 711.00         |
| AAA.1  | 17       | 29.17            | 491.59         | 1125.22          | 4.55             | 722.00         |
| AAA.1  | 18       | 29.16            | 490.96         | 1124.84          | 4.96             | 704.00         |
| AAA.1  | 19       | 29.17            | 491.82         | 1125.40          | 5.38             | 722.00         |
| AAA.1  | 20       | 29.16            | 490.59         | 1124.88          | 345.14           | 8849.00        |
| AAA.1  | 21       | 29.17            | 491.57         | 1125.12          | 291.88           | 6502.00        |
| AAA.1  | 22       | 29.17            | 491.82         | 1125.26          | 5.04             | 715.00         |
| AAA.1  | 23       | 29.17            | 491.02         | 1124.99          | 5.15             | 721.00         |
| AAA.1  | 24       | 29.16            | 491.70         | 1124.83          | 4.80             | 727.00         |
| AAA.1  | 25       | 29.17            | 492.09         | 1124.78          | 5.75             | 717.00         |
| AAB.1  | 1        | 29.16            | 491.87         | 1124.84          | 5.14             | 717.00         |
| AAB.1  | 2        | 29.17            | 491.70         | 1125.08          | 5.41             | 711.00         |
| AAB.1  | 3        | 29.17            | 491.16         | 1125.09          | 4.61             | 726.00         |
| AAB.1  | 4        | 29.17            | 491.74         | 1124.55          | 4.73             | 721.00         |
| AAB.1  | 5        | 29.17            | 491.64         | 1125.14          | 5.94             | 718.00         |
| AAB.1  | 6        | 29.17            | 490.39         | 1125.31          | 5.08             | 722.00         |
| AAB.1  | 7        | 29.17            | 491.73         | 1125.01          | 6.16             | 710.00         |
| AAB.1  | 8        | 29.16            | 490.74         | 1125.63          | 4.71             | 715.00         |
| AAB.1  | 9        | 29.17            | 491.70         | 1125.45          | 4.69             | 715.00         |

Data table:

1. 공정 split 평가 ...

4. Visualization

Max(reflect\_power) for, Avg(rf\_power) for, Avg(precursor\_flow\_rate) fo...



Data Relationships (Linear Regression)

| Y (numerical)  | X (numerical)    | p-value ▲ | FStat     | RSq ▼ | R     | Df  |
|----------------|------------------|-----------|-----------|-------|-------|-----|
| Sum(value) for | Max(reflect_p... | 0.00E+000 | 125351.35 | 0.99  | 1.00  | 998 |
| Sum(value) for | Avg(rf_power)... | 5.13E-001 | 0.43      | 0.00  | 0.02  | 998 |
| Sum(value) for | Avg(precursor... | 6.37E-001 | 0.22      | 0.00  | -0.01 | 998 |
| Sum(value) for | Avg(chamber_...  | 8.67E-001 | 0.03      | 0.00  | -0.01 | 998 |

Data Relationships (Details)

